

## Semi-prime H-numbers

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### Description

This problem is based on an exercise of David Hilbert, who pedagogically suggested that one study the theory of

 $4n+1$ 

numbers. Here, we do only a bit of that.

An **H**-number is a positive number which is one more than a multiple of four: 1, 5, 9, 13, 17, 21,... are the **H**-numbers. For this problem we pretend that these are the *only* numbers. The **H**-numbers are closed under multiplication.

As with regular integers, we partition the **H**-numbers into units, **H**-primes, and **H**-composites. 1 is the only unit. An **H**-number **h** is **H**-prime if it is not the unit, and is the product of two **H**-numbers in only one way:  $1 \times h$ . The rest of the numbers are **H**-composite.

For examples, the first few **H**-composites are:  $5 \times 5 = 25$ ,  $5 \times 9 = 45$ ,  $5 \times 13 = 65$ ,  $9 \times 9 = 81$ ,  $5 \times 17 = 85$ .

Your task is to count the number of **H**-semi-primes. An **H**-semi-prime is an **H**-number which is the product of exactly two **H**-primes. The two **H**-primes may be equal or different. In the example above, all five numbers are **H**-semi-primes.  $125 = 5 \times 5 \times 5$  is not an **H**-semi-prime, because it's the product of three **H**-primes.

Each line of input contains an **H**-number  $\leq 1,000,001$ . The last line of input contains 0 and this line should not be processed.

For each inputted **H**-number **h**, print a line stating **h** and the number of **H**-semi-primes between 1 and **h** inclusive, separated by one space in the format shown in the sample.

### Input

Please Input Input Here

### Output

Please Input Output Here

### Samples

input

Copy

|        |      |
|--------|------|
| 21     |      |
| 85     |      |
| 789    |      |
| 0      |      |
| output | Copy |
| 21 0   |      |
| 85 5   |      |
| 789 62 |      |

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HZNUOJ is based on [HUSTOJ](#)

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